

Question 1: (25 marks)

- a) Solve the quadratic equation $x^2 + 4x + 5 = 0$, giving your answers in the form $a + ib$. 2
- b) Form the quadratic equation with roots $1 + 3i$ and $1 - 3i$. 2
- c) If $z = 3 - 2i$ and $w = 2 + i$, find, expressing in the form $a + ib$:
- i) $3z - 2w$ 2
 - ii) iw 2
 - iii) $\frac{w}{z}$ 2
 - iv) $w^2 - z^2$ 2
- d) i) Express $\sqrt{8 + 6i}$ in the form $a + ib$. 4
- ii) Hence solve the equation: $z^2 + 2(1 + 2i)z - (11 + 2i) = 0$ 3
- e) i) Express $\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}$ in mod-arg form. 2
- ii) Use De Moivre's theorem to simplify $\left(\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right)^{60}$, 2
- expressing your answer in the form $a + ib$
- f) Evaluate i^{2009} . 2

Question 2: (25 marks)

- a) i) Simplify $\left(\text{cis}\frac{\pi}{3}\right)\left(\text{cis}\frac{\pi}{4}\right)$. Express your answer in the form $a + ib$. 3
- ii) Hence evaluate $\cos\frac{7\pi}{12}$ in surd form. 2
- b) i) Use De Moivre's theorem to express $\cos 4\theta$ in terms of $\sin\theta$ and $\cos\theta$. 3
- ii) Express $\sin 4\theta$ in terms of $\sin\theta$ and $\cos\theta$. 1
- iii) Hence show that $\tan 4\theta = \frac{4t - 4t^3}{1 - 6t^2 + t^4}$ where $t = \tan\theta$. 3
- iv) Using this result or otherwise, solve the equation $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$. 3
- c) i) If $z = \cos\theta + i\sin\theta$, use de Moivre's Theorem to show that $z^n + \frac{1}{z^n} = 2\cos n\theta$ 1
- ii) Find an expression for $\cos^4\theta$ in terms of multiples of $\cos\theta$. 3
- d) i) Solve the equation $z^6 + 1 = 0$, giving the roots in the form $a + ib$. 3
- ii) Hence factorise $z^6 + 1$ into real quadratic factors. 3

Question 3: (25 marks)

- a) One vertex of an equilateral triangle OAB is $A(1+i)$. If O is the origin, find the coordinates of B if it lies in the second quadrant. 3
- b) Given O is the origin and A is represented by the complex number $\sqrt{3}+i$:
- find C if OA is rotated through $\left(-\frac{\pi}{3}\right)$ and doubled in length to form C ; 2
 - find the point B which forms the parallelogram $OABC$ 2
- c) Prove that $|z_1 - z_2|^2 + |z_1 + z_2|^2 = 2|z_1|^2 + 2|z_2|^2$ 4
- d) If w is the complex cube root of unity with the smallest positive argument:
- show that the other complex cube root is w^2 2
 - prove that $1+w+w^2=0$ 2
 - evaluate $(1+3w+w^2)(1+w+3w^2)$ 1
 - find the quadratic equation which has the roots $(1+3w+w^2)$ and $(1+w+3w^2)$ 2
- e) If $z = \frac{1+i}{1-i}$ and $w = \frac{2}{1-\sqrt{3}i}$:
- express z and w in modulus argument form; 4
 - plot $z, w, z+w$ on an Argand diagram; 1
 - show that $\tan \frac{5\pi}{12} = \sqrt{3}+2$ 2

END OF TEST

EXTENSION 2 MATHEMATICS TASK 2 SOLUTIONS

Question 1:

- a) $x^2 + 4x + 5 = 0$
- $$x = \frac{-4 \pm \sqrt{16-20}}{2}$$
- $$= \frac{-4 \pm \sqrt{-4}}{2}$$
- $$= \frac{-4 \pm 2i}{2}$$
- $$= -2 \pm i$$
- b) $\alpha + \beta = (1+3i) + (1-3i)$
- $$= 2$$
- $$\alpha\beta = (1+3i)(1-3i)$$
- $$= 1-9i^2$$
- $$= 10$$
- $$x^2 - 2x + 10 = 0$$
- c) i) $3z - 2w = 3(3-2i) - 2(2+i)$
- $$= 9-6i-4-2i$$
- $$= 5-8i$$
- ii) $iw = i(2+i)$
- $$= 2i+i^2$$
- $$= -1+2i$$
- iii) $\frac{w}{z} = \frac{2+i}{3-2i} \times \frac{3+2i}{3+2i}$
- $$= \frac{(2+i)(3+2i)}{(3-2i)(3+2i)}$$
- $$= \frac{6+7i+2i^2}{9-4i^2}$$
- $$= \frac{4}{13} + i\frac{7}{13}$$
- iv) $w^2 - z^2 = (2+i)^2 - (3-2i)^2$
- $$= (4+4i+i^2) - (9-12i+4i^2)$$
- $$= (3+4i) - (5-12i)$$
- $$= -2+16i$$

$$d) \quad i) \sqrt{8+6i} = a+ib$$

$$8+6i = (a+ib)^2$$

$$= a^2 - b^2 + i2ab$$

$$a^2 - b^2 = 8 \quad (1)$$

$$2ab = 6 \quad (2)$$

$$\text{Using } (a^2 + b^2)^2 = (a^2 - b^2)^2 + (2ab)^2$$

$$= 8^2 + 6^2$$

$$= 100$$

$$a^2 + b^2 = 10 \quad (\text{as } a^2 + b^2 > 0) \quad (3)$$

$$(1) + (3)$$

$$2a^2 = 18$$

$$a^2 = 9$$

$$a = \pm 3$$

$$\text{When } a = 3, b = 1$$

$$\text{When } a = -3, b = -1$$

$$\therefore \sqrt{8+6i} = \pm(3+i)$$

$$ii) \quad z^2 + 2(1+2i)z - (11+2i) = 0$$

$$z^2 + 2(1+2i)z = 11+2i$$

$$z^2 + 2(1+2i)z + (1+2i)^2 = 11+2i + (1+2i)^2$$

$$(z + [1+2i])^2 = 11+2i + 1 + 4i + 4i^2$$

$$= 12 + 6i - 4$$

$$= 8 + 6i$$

$$z + 1 + 2i = \pm \sqrt{8+6i}$$

$$= \pm(3+i)$$

$$z + 1 + 2i = 3 + i$$

$$z = 2 - i$$

$$z + 1 + 2i = -3 - i$$

$$z = -4 - 3i$$

$$e) \quad i) \quad r = \sqrt{\left(\frac{\sqrt{2}}{2}\right)^2 + \left(\frac{-\sqrt{2}}{2}\right)^2}$$

$$= 1$$

$$\theta = \tan^{-1}(1) \text{ in } 4^{\text{th}} \text{ quadrant}$$

$$= -\frac{\pi}{4}$$

$$\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} = \text{cis}\left(-\frac{\pi}{4}\right)$$

$$ii) \quad \left(\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right)^{60} = \left(\text{cis}\left[-\frac{\pi}{4}\right]\right)^{60}$$

$$= \left(\cos\left[-\frac{\pi}{4}\right] + i\sin\left[-\frac{\pi}{4}\right]\right)^{60}$$

$$= \cos(-15\pi) + i\sin(-15\pi)$$

$$= \cos 15\pi - i\sin 15\pi$$

$$= -1$$

$$f) \quad i^{2009} = (i^4)^{502} \times i$$

$$= 1 \times i$$

$$= i$$

Question 2:

$$\begin{aligned} \text{a) i) } \left(\operatorname{cis} \frac{\pi}{3} \right) \times \left(\operatorname{cis} \frac{\pi}{4} \right) &= \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \\ &= \left(\frac{1}{2} + i \frac{\sqrt{3}}{2} \right) \left(\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}} \right) \\ &= \frac{1}{2} (1 + i\sqrt{3}) \frac{1}{\sqrt{2}} (1 + i) \\ &= \frac{1}{2\sqrt{2}} (1 + i + i\sqrt{3} + i^2 \sqrt{3}) \\ &= \frac{\sqrt{2}}{4} (1 - \sqrt{3}) + i \frac{\sqrt{2}}{4} (1 + \sqrt{3}) \end{aligned}$$

$$\begin{aligned} \text{ii) } \left(\operatorname{cis} \frac{\pi}{3} \right) \left(\operatorname{cis} \frac{\pi}{4} \right) &= \operatorname{cis} \left(\frac{\pi}{3} + \frac{\pi}{4} \right) \\ &= \operatorname{cis} \frac{7\pi}{12} \end{aligned}$$

Equating real parts:

$$\cos \frac{7\pi}{12} = \frac{\sqrt{2}(1 - \sqrt{3})}{4}$$

$$\text{b) i) } (\cos \theta + i \sin \theta)^4 = \cos 4\theta + i \sin 4\theta$$

$$\begin{aligned} (\cos \theta + i \sin \theta)^4 &= \cos^4 \theta + 4 \cos^3 \theta (i \sin \theta) + 6 \cos^2 \theta (i \sin \theta)^2 + 4 \cos \theta (i \sin \theta)^3 + (i \sin \theta)^4 \\ &= \cos^4 \theta + i 4 \cos^3 \theta \sin \theta + i^2 6 \cos^2 \theta \sin^2 \theta + i^3 4 \cos \theta \sin^3 \theta + i^4 \sin^4 \theta \\ &= (\cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta) + i (4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta) \\ \cos 4\theta &= \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta \end{aligned}$$

$$\text{ii) } \sin 4\theta = 4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta$$

$$\begin{aligned} \text{iii) } \tan 4\theta &= \frac{\sin 4\theta}{\cos 4\theta} \\ &= \frac{4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta}{\cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta} \\ &= \frac{4 \cos^3 \theta \sin \theta}{\cos^4 \theta} - \frac{4 \cos \theta \sin^3 \theta}{\cos^4 \theta} \\ &= \frac{\cos^4 \theta}{\cos^4 \theta} - \frac{6 \cos^2 \theta \sin^2 \theta}{\cos^4 \theta} + \frac{\sin^4 \theta}{\cos^4 \theta} \\ &= \frac{4t - 4t^3}{1 - 6t^2 + t^4} \quad \text{where } t = \tan \theta \end{aligned}$$

$$\text{iv) } x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$$

$$4x - 4x^3 = 1 - 6x^2 + x^4$$

$$\frac{4x - 4x^3}{1 - 6x^2 + x^4} = 1$$

$$\text{Let } x = \tan \theta = t:$$

$$\frac{4t - 4t^3}{1 - 6t^2 + t^4} = 1$$

$$\tan 4\theta = 1$$

$$4\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4}$$

$$\theta = \frac{\pi}{16}, \frac{5\pi}{16}, \frac{9\pi}{16}, \frac{13\pi}{16}$$

$$x = \tan \frac{\pi}{16}, \tan \frac{5\pi}{16}, \tan \frac{9\pi}{16}, \tan \frac{13\pi}{16}$$

$$\text{c) i) } z^n + \frac{1}{z^n} = z^n + z^{-n}$$

$$\begin{aligned} &= (\cos \theta + i \sin \theta)^n + (\cos \theta + i \sin \theta)^{-n} \\ &= \cos n\theta + i \sin n\theta + \cos(-n\theta) + i \sin(-n\theta) \\ &= \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta \\ &= 2 \cos n\theta \end{aligned}$$

$$\text{ii) Let } n = 1:$$

$$z + \frac{1}{z} = 2 \cos \theta$$

$$\left(z + \frac{1}{z} \right)^4 = 16 \cos^4 \theta$$

$$\begin{aligned} 16 \cos^4 \theta &= z^4 + 4z^2 + 6 + \frac{4}{z^2} + \frac{1}{z^4} \\ &= \left(z^4 + \frac{1}{z^4} \right) + 4 \left(z^2 + \frac{1}{z^2} \right) + 6 \\ &= 2 \cos 4\theta + 8 \cos 2\theta + 6 \\ \cos^4 \theta &= \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8} \end{aligned}$$

d) i) $z^6 = -1$

$$= \text{cis} \pi$$

$$z = (\text{cis} \pi)^{\frac{1}{6}}$$

$$= \text{cis} \left(\frac{\pi + 2k\pi}{6} \right) \text{ where } k = 0, 1, 2, 3, 4, 5$$

$$z_0 = \text{cis} \frac{\pi}{6} = \frac{\sqrt{3}}{2} + i \frac{1}{2}$$

$$z_1 = \text{cis} \frac{\pi}{2} = i$$

$$z_2 = \text{cis} \frac{5\pi}{6} = -\frac{\sqrt{3}}{2} + i \frac{1}{2}$$

$$z_3 = \text{cis} \frac{7\pi}{6} = -\frac{\sqrt{3}}{2} - i \frac{1}{2}$$

$$z_4 = \text{cis} \frac{3\pi}{2} = -i$$

$$z_5 = \text{cis} \frac{11\pi}{6} = \frac{\sqrt{3}}{2} - i \frac{1}{2}$$

ii) Note:

$$z_3 = \text{cis} \frac{7\pi}{6} = \text{cis} \left(-\frac{5\pi}{6} \right) = \overline{z_2}$$

$$z_5 = \text{cis} \frac{11\pi}{6} = \text{cis} \left(-\frac{\pi}{6} \right) = \overline{z_0}$$

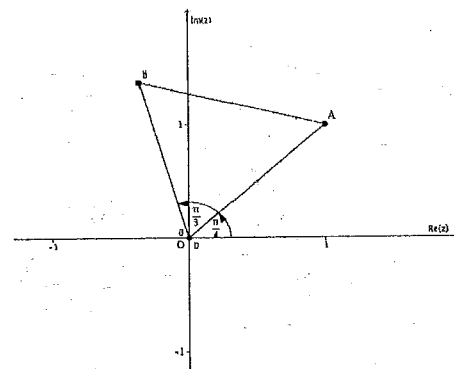
$$\begin{aligned} z + \overline{z} &= \cos \theta + i \sin \theta + \cos \theta - i \sin \theta \\ &= 2 \cos \theta \\ &= 2 \text{Re}(z) \end{aligned}$$

$$\begin{aligned} z \overline{z} &= (\cos \theta + i \sin \theta)(\cos \theta - i \sin \theta) \\ &= \cos^2 \theta - i^2 \sin^2 \theta \\ &= \cos^2 \theta + \sin^2 \theta \\ &= 1 \end{aligned}$$

$$\begin{aligned} z^6 + 1 &= (z - i)(z + i)(z - z_0)(z - \overline{z_0})(z - z_2)(z - \overline{z_2}) \\ &= (z^2 - i^2)(z^2 - [z_0 + \overline{z_0}]z + z_0 \overline{z_0})(z^2 - [z_2 + \overline{z_2}]z + z_2 \overline{z_2}) \\ &= (z^2 + 1)(z^2 - \sqrt{3}z + 1)(z^2 + \sqrt{3}z + 1) \end{aligned}$$

Question 3:

a)



$$\begin{aligned} \left(\text{cis} \frac{\pi}{3} \right) (1+i) &= \left(\frac{1}{2} + i \frac{\sqrt{3}}{2} \right) (1+i) \\ &= \frac{1}{2} + \frac{1}{2}i + i \frac{\sqrt{3}}{2} + i^2 \frac{\sqrt{3}}{2} \\ &= \left(\frac{1}{2} - \frac{\sqrt{3}}{2} \right) + i \left(\frac{1}{2} + \frac{\sqrt{3}}{2} \right) \end{aligned}$$

Coordinates of B: $\left(\frac{1}{2} - \frac{\sqrt{3}}{2}, \frac{1}{2} + \frac{\sqrt{3}}{2} \right)$

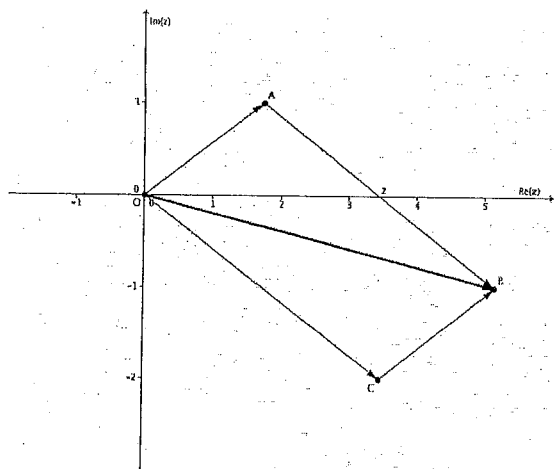
b)

$$\begin{aligned} \text{i) } |OA| &= \sqrt{(\sqrt{3})^2 + 1^2} \\ &= 2 \\ |OC| &= 2 \times |OA| \\ &= 4 \end{aligned}$$

$$\begin{aligned} 2 \text{cis} \left(-\frac{\pi}{3} \right) (\sqrt{3} + i) &= 2 \left(\frac{1}{2} - i \frac{\sqrt{3}}{2} \right) (\sqrt{3} + i) \\ &= (1 - i\sqrt{3})(\sqrt{3} + i) \\ &= \sqrt{3} + i - 3i - i^2 \sqrt{3} \\ &= 2\sqrt{3} - 2i \end{aligned}$$

Coordinates of C: $(2\sqrt{3}, -2)$

ii)



$$\begin{aligned}\overrightarrow{OC} &= \overrightarrow{OA} + \overrightarrow{OB} \\ &= (\sqrt{3} + i) + (2\sqrt{3} - 2i) \\ &= 3\sqrt{3} - i\end{aligned}$$

Coordinates of C: $(3\sqrt{3}, -1)$

$$\begin{aligned}\text{c) } |z_1 - z_2|^2 + |z_1 + z_2|^2 &= (z_1 - z_2)(\overline{z_1 - z_2}) + (z_1 + z_2)(\overline{z_1 + z_2}) \\ &= (z_1 - z_2)(\overline{z_1} - \overline{z_2}) + (z_1 + z_2)(\overline{z_1} + \overline{z_2}) \\ &= z_1\overline{z_1} - z_1\overline{z_2} - z_2\overline{z_1} + z_2\overline{z_2} + z_1\overline{z_1} + z_1\overline{z_2} + z_2\overline{z_1} + z_2\overline{z_2} \\ &= 2z_1\overline{z_1} + 2z_2\overline{z_2} \\ &= 2|z_1|^2 + 2|z_2|^2\end{aligned}$$

$$\begin{aligned}\text{d) } i) z^3 &= 1 \\ &= \text{cis } 0 \\ z &= (\text{cis } 0)^{\frac{1}{3}} \\ &= \text{cis} \left(\frac{0 + 2k\pi}{3} \right) \text{ where } k = 0, 1, 2\end{aligned}$$

$$\begin{aligned}z_0 &= \text{cis } 0 \\ &= 1\end{aligned}$$

$$\begin{aligned}z_1 &= \text{cis} \frac{2\pi}{3} \\ &= -\frac{1}{2} + i\frac{\sqrt{3}}{2}\end{aligned}$$

$$\begin{aligned}z_2 &= \text{cis} \frac{4\pi}{3} \\ &= -\frac{1}{2} - i\frac{\sqrt{3}}{2}\end{aligned}$$

$$\text{Let } w = \text{cis} \frac{2\pi}{3} :$$

$$\begin{aligned}w^2 &= \left(\text{cis} \frac{2\pi}{3} \right)^2 \\ &= \text{cis} \frac{4\pi}{3} \\ &= z_2\end{aligned}$$

$$\begin{aligned}\text{ii) } 1 + w + w^2 &= \frac{1(w^3 - 1)}{w - 1} \\ &= \frac{1 - 1}{w - 1} \quad (\text{as } w^3 = 1) \\ &= 0\end{aligned}$$

$$\begin{aligned}\text{iii) } (1 + 3w + w^2)(1 + w + 3w^2) &= ([1 + w^2] + 3w)([1 + w] + 3w^2) \\ &= (-w + 3w)(-w^2 + 3w^2) \\ &= 2w \times 2w^2 \\ &= 4w^3 \\ &= 4\end{aligned}$$

$$\begin{aligned}\text{iii) } (1 + 3w + w^2) + (1 + w + 3w^2) &= 2w + 2w^2 \\ &= 2w(1 + w) \\ &= 2w(-w^2) \\ &= -2w^3 \\ &= -2\end{aligned}$$

$$(1 + 3w + w^2)(1 + w + 3w^2) = 4$$

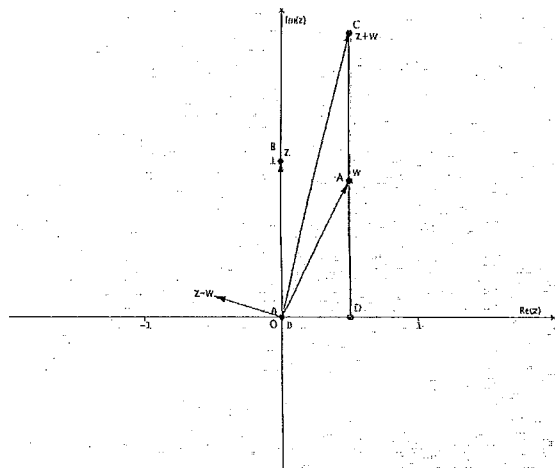
$$x^2 + 2x + 4 = 0$$

e)

$$\begin{aligned} \text{i) } z &= \frac{\sqrt{2} \operatorname{cis} \frac{\pi}{4}}{\sqrt{2} \operatorname{cis} \left(-\frac{\pi}{4} \right)} \\ &= \operatorname{cis} \frac{\pi}{2} \end{aligned}$$

$$\begin{aligned} w &= \frac{2 \operatorname{cis} 0}{2 \operatorname{cis} \left(-\frac{\pi}{3} \right)} \\ &= \operatorname{cis} \frac{\pi}{3} \end{aligned}$$

ii)



$$\begin{aligned} \angle AOD &= \arg(w) \\ &= \frac{\pi}{3} \end{aligned}$$

$$\angle BOA = \frac{\pi}{6} \quad (\text{adjacent complementary } \angle s)$$

As $|z| = |w|$, $OBCA$ is a rhombus.

So diagonals bisect the angles through which they pass.

$$\begin{aligned} \angle COA &= \frac{\pi}{6} \div 2 \\ &= \frac{\pi}{12} \end{aligned}$$

$$\begin{aligned} \angle COD &= \frac{\pi}{3} + \frac{\pi}{12} \\ &= \frac{5\pi}{12} \end{aligned}$$

In $\triangle AOD$:

$$\sin \angle AOD = \frac{DA}{1}$$

$$\sin \frac{\pi}{3} = DA$$

$$DA = \frac{\sqrt{3}}{2}$$

$$\cos \angle AOD = \frac{DO}{1}$$

$$\cos \frac{\pi}{3} = DO$$

$$DO = \frac{1}{2}$$

In $\triangle COD$:

$$\begin{aligned} \tan \frac{5\pi}{12} &= \frac{\frac{\sqrt{3}}{2} + 1}{\frac{1}{2}} \\ &= \frac{\frac{\sqrt{3} + 2}{2}}{\frac{1}{2}} \\ &= \sqrt{3} + 2 \end{aligned}$$